

Quantum Fault Tolerance Meets Toom's Contours

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1 Hadamard and S Gates

Claim 1.1. For any generator subset S' of size d' , denote by $X_{S'}$ and $Z_{S'}$ the following assignments:

$$X_{S'} := \sum_{u \in \langle S' \rangle} \theta_{u, S'} \quad Z_{S'} := \sum_{u \in \langle S/S' \rangle} \theta_{u, S'}$$

Then $X_{S'}$ and $Z_{S'}$ are binary representations of anticommuting logical codewords of the (d, d') -Toric code. In particular, $X_{S'} \in \mathcal{C}_X / \mathcal{C}_Z^\perp$, $Z_{S'} \in \mathcal{C}_Z / \mathcal{C}_X^\perp$, and $X_{S'} Z_{S'} = 1$.

Proof. By definition $X_{S'}$ and $Z_{S'}$ are binary assignments on the d' -dimensional faces. We start by showing that $X_{S'} \in \mathcal{C}_X$ and $Z_{S'} \in \mathcal{C}_Z$:

$$\begin{aligned} \partial^- \sum_{u \in \langle S' \rangle} \theta_{u, S'} &= \sum_{u \in \langle S' \rangle} \sum_{g \in S'} \theta_{u, S'/g} + \theta_{\mathbf{g}u, S'/g} \\ &= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{u, S'/g} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{g}u, S'/g} \\ &= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{u, S'/g} + \sum_{u \in \langle S' \rangle} \theta_{u, S'/g} = 0 \\ \partial^+ \sum_{u \in \langle S' \rangle} \theta_{u, S/S'} &= \sum_{u \in \langle S' \rangle} \sum_{g \in S'} \theta_{u, S' \cup g} + \theta_{\mathbf{g}^{-1}u, S/S' \cup g} \\ &= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{u, S' \cup g} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{g}^{-1}u, S/S' \cup g} \\ &= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{u, S' \cup g} + \sum_{u \in \langle S' \rangle} \theta_{u, S' \cup g} = 0 \end{aligned}$$

Thus $X_{S'} \in \mathcal{C}_X$ and $Z_{S'} \in \mathcal{C}_Z$, now for showing that they are not binary representations of stabilizers, it's enough to show they admit no vanishing product.

$$X_{S'} Z_{S'} = \sum_{u \in \langle S' \rangle} \theta_{u, S'} \sum_{u \in \langle S/S' \rangle} \theta_{u, S'} = \theta_{\mathbf{e}, S'} \theta_{\mathbf{e}, S'} = 1$$

□

[COMMENT] I should write the action such the map act only on the rooted vertex: $\phi(\theta_{\mathbf{v}, \mathbf{S}'}) \rightarrow \theta_{\phi(\mathbf{v}), \mathbf{S}'}$. And then it works.

Definition 1.2 (The $g_a \leftrightarrow g_b$ Transport). Let d, d' be integers, $\mathcal{G}$ be a (d, d') -Toric complex, generated by generator set S . Since the vertices in $\mathcal{G}$ are the elements of the abelian group generated by S , each of vertex $v \in \mathcal{G}$ has a unique presentation of:

$$v = \prod_{i \in [d]} (g_i)^{x_i}$$

For generators g_a, g_b in S , we define the $g_a \leftrightarrow g_b$ transport map: $\phi_{g_a \leftrightarrow g_b} : \mathcal{G} \rightarrow \mathcal{G}$ as:

$$\phi(v)_{g_a \leftrightarrow g_b} := \prod_{i \in [d] \setminus \{a, b\}} (g_i)^{x_i} g_a^{x_b} g_b^{x_a}$$

$$\phi_{g_a \leftrightarrow g_b}(vu) = \phi_{g_a \leftrightarrow g_b}\left(\prod_{i \in [d]} (g_i)^{x_i + y_i}\right) = \prod_{i \in [d] \setminus \{a, b\}} (g_i)^{x_i + y_i} g_a^{x_b + y_b} g_b^{x_a + y_a} = \phi_{g_a \leftrightarrow g_b}(v) \phi_{g_a \leftrightarrow g_b}(u)$$

$$\begin{aligned} \phi(\theta_{\mathbf{v}, \mathbf{S}'})_{g_a \leftrightarrow g_b} &= \phi\left(\bigcup_{I \subset \mathbf{S}'} \left\{ \left(\prod_{g \in I} g \right) v \right\}\right)_{g_a \leftrightarrow g_b} \\ &= \left(\bigcup_{I \subset \mathbf{S}'} \left\{ \phi\left(\left(\prod_{g \in I} g \right) v \right)_{g_a \leftrightarrow g_b} \right\} \right) \\ &= \left(\bigcup_{I \subset \mathbf{S}'} \left\{ \phi\left(\left(\prod_{g \in I} g \right) v \right)_{g_a \leftrightarrow g_b} \right\} \right) \\ &= \theta_{\phi(v)_{g_a \leftrightarrow g_b}, \phi(\mathbf{S}')_{g_a \leftrightarrow g_b}} \end{aligned}$$

So the map sends bits to bits and checks to checks.

Claim 1.3. For $d' = d/2$, the H gate, can be computed transversally.

Proof. The action of the permutation $g_a \leftrightarrow g_b$, $v \mapsto$, sends checks to checks.

$$\theta_{\mathbf{v}, \mathbf{S}'} \rightarrow \theta_{\mathbf{v}, \phi(\mathbf{S}')}$$

And:

$$\begin{aligned} \partial^+ \phi(\theta_{\mathbf{v}, \mathbf{S}'}) &= \partial^+ \theta_{\phi(\mathbf{v}), \mathbf{S}'} = \sum_{g \in S/S'} \theta_{\phi(\mathbf{v}), \mathbf{S}' \cup g} + \sum_{g \in S/S'} \theta_{\mathbf{g}^{-1} \phi(\mathbf{v}), \mathbf{S}' \cup g} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \phi(\cdot) &= \phi\left(\sum_{u \in \langle \mathbf{S}' \rangle} \theta_{\mathbf{u}\mathbf{v}, \mathbf{S}'}\right) \\ &= \sum_{u \in \langle \mathbf{S}' \rangle} \theta_{\phi(\mathbf{u}\mathbf{v}), \mathbf{S}'} = \sum_{u \in \langle \phi(\mathbf{S}') \rangle} \theta_{\mathbf{u}\phi(\mathbf{v}), \mathbf{S}'} \end{aligned}$$

$$z\phi(z) = \sum_{u \in \langle S' \rangle} \theta_{\mathbf{uv}, S'} \theta_{\phi^{-1}(\mathbf{uv}), S'}$$

And there is only one vertex in $\langle S' \rangle v$ for which $\phi(uv) = uv$. But this is not $H^{\otimes n}$. Required condition:

$$\begin{aligned} |\phi(S) \cap S/S'| &\geq 1 \\ g_1^x g_2^y &= g_2^x g_3^y \rightarrow x, y = 0 \end{aligned}$$

It seems that if ϕ is a cycle $g_i \rightarrow g_{i+1}$ then it is fine. The prove by induction on $|S'|$, after we throw away the first generator (in the example $g_3 \rightarrow y = 0$), then one can assume that $S' \leftarrow S'/g_2$. $\square$

Claim 1.4. The S gate can be computed transversally.

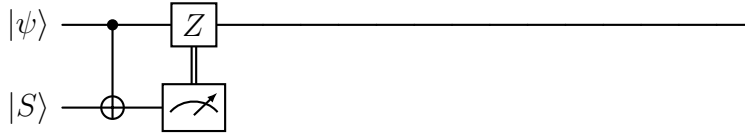


Figure 1: Simulating the S -gate, using the $|S\rangle$ magic state.

2 Toric Encoded Circuits

Definition 2.1 (Troic Encoded Circuit). Let C be a quantum circuit. We say that C is *Toric encoded* if the following holds:

1. There is a Toric code $\mathcal{C}$ such that the qubits of $\mathcal{C}$ can be partitioned into registers $R_1, \dots, R_k$, and each of the registers is encoded by $\mathcal{C}$.
2. The gates in C with even time coordinates are sweep decoding gates.

For a set of faults $\mathcal{E}$ the gates in $C[\mathcal{E}]$ can be decomposed to cycles, of the following form:

$$C = D_l P_l \Lambda_l, \dots, D_1 P_1 \Lambda_1$$

Definition 2.2 (Base graph G_o). Let C be a quantum circuit, at depth d , with registers $R_1, \dots, R_k$, all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by $\mathcal{G}_1, \dots, \mathcal{G}_k$ the Toric complexes that induce the codes. We define the base graph of C , denoted by $G_o = (V_o, E_o)$ as follows:

1. The vertices V_o are tuples, where the first coordinates correspond to a vertex in the union over the vertices in $\mathcal{G}_1, \dots, \mathcal{G}_k$, and the second coordinate is associated with a time. Namely:

$$V_o := \bigsqcup_i V(\mathcal{G}_i) \times [d]$$

2. The edges E_o are derived from the original edges in $\mathcal{G}_1, \dots, \mathcal{G}_k$, with the addition that any two vertices in preceding times t and $t+1$ that support qubits (faces) such that C acts non-trivially on those qubits at time t are also connected by an edge. Namely:

$$E_o := \{ \{(v, t), (u, t')\} : \text{if} \\ \text{either there exists } i \text{ such } \{v, u\} \in \mathcal{G}_i \text{ and } t' = t, t+1 \\ \text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1} + 1 \}$$

[COMMENT] Here we should add that a vertex v belongs to location l if there is a qubit f that v belongs to its associated face.

Definition 2.3 (Time Graph of $C(\mathcal{E})$). For a quantum circuit C and set of faults $\mathcal{E}$, let us denote the base graph of $C[\mathcal{E}]$ by G_o . For each time t , denote the deviation induced by $\mathcal{E}$ at time t by $\mathcal{D}_t$, so $\partial\mathcal{D}_t$ is the syndrome that the qubits in the deviation admit. We define the history graph $\mathcal{H} = (V_{\mathcal{H}}, E_{\mathcal{H}})$ as follows:

1. The vertices $V_{\mathcal{H}}$ are taken to be the vertices of G_o , namely tuples $(v, t) \in V_o$, that satisfy that v belongs to a face associated with a check, which, at time t , belongs to the $\partial\mathcal{D}_t$. Namely:

$$\{v_t : v_t = (v, t) \in V_o \text{ and there exists a check } f \in C \text{ such } v \in f \in \partial\mathcal{D}_t\}$$

2. The edges $E_{\mathcal{H}}$ is partitioned into two types $A_{\mathcal{H}}$ and $F_{\mathcal{H}}$, where $A_{\mathcal{H}}$ are the edges in E_o restricted to $V_{\mathcal{H}}$ and $F_{\mathcal{H}}$ are defined to be the . Namely:

$$A_{\mathcal{H}} = \{e = \{u, v\} \in E_o : u, v \in V_{\mathcal{H}} \text{ and } T(v) = T(u) \pm 1\} \\ F_{\mathcal{H}} = \{e = \{u, v\} \in E_o : \text{exists } w \in V_{\mathcal{H}} \text{ such } \{v, w\}, \{u, w\} \in A_{\mathcal{H}}\}$$

By definition of the construction, $G_{\mathcal{H}}$ is a time-graph.

Definition 2.4 (**Size()**-Function). Let S be a subset of vertices, we define the *size* of S to be:

$$\mathbf{Size}(S) := \sum_i^{d+1} \max_{v \in S} L_i(v)$$

Claim 2.5. Let $\mathbf{S} = \{S_1, S_2, \dots, S_k\}$ where $S_i \subset \mathbb{R}^d$, introduce an edge between S_i and S_j iff $S_i \cap S_j \neq \emptyset$. Assume that $\mathbf{S}$ is connected. Then $\mathbf{Size}(\cup_{i=1}^k S_i) \leq \sum_{i=1}^k \mathbf{Size}(S_i)$.

Proof. It suffices to prove that if $R_1 \cap R_2 \neq \emptyset$, then $\mathbf{Size}(R_1 \cup R_2) \leq \mathbf{Size}(R_1) + \mathbf{Size}(R_2)$. Let $a_{i,j} = \max_{v \in R_j} L_i(v)$. Then for any $w \in R_1 \cap R_2$ we have:

$$\begin{aligned} \mathbf{Size}(R_1 \cup R_2) &= \sum_i^{d+1} \max a_{i,1}, a_{i,2} = \sum_i^{d+1} a_{i,1} + a_{i,2} - \min a_{i,1}, a_{i,2} \\ &\leq \sum_i^{d+1} a_{i,1} + a_{i,2} - L_i(w) = \mathbf{Size}(R_1) + \mathbf{Size}(R_2) - \mathbf{Size}(w) \\ &= \mathbf{Size}(R_1) + \mathbf{Size}(R_2) \end{aligned}$$

□

Definition 2.6 (Size Invariant Action). [COMMENT] Here we classify both transversal gates, and rotation on the Toric. An action g is be said size invariant action if, for any assignment z , it holds that $\mathbf{Size}(gz) = \mathbf{Size}(z)$.

Claim 2.7. The sweep rule is Size ξ -shrinking.

Proof.

□

[COMMENT] Eventually, we might not used the projected graph, and then we will split to history of the slices lemma for 3 parts. The first is when time = $3t$, in that we apply an invariant action so.